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Multi-level uncertainty aware learning for semi-supervised dental panoramic caries segmentation

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ABSTRACT

Dental caries segmentation based on oral panoramic medical images is a demanding medical task. However, it is challenging to determine imaging diagnosis results as the actual pathological changes under certain conditions, especially in the suspected pathological locations that distinguish the artifacts from the lesions. This paper proposes introducing the semi-supervised learning framework to redistribute the collected cases to set up exact lesion areas and divide suspected areas with uncertainties into unlabeled data instead of reducing dataset labeling costs. Consistent regularization learning allows us to exploit the feature of ambiguous lesions to optimize decision boundaries. Unfortunately, the labels generated by the general SSL method for fuzzy regions are volatile, especially encountering no specific morphological rules and various scales that fluctuate more seriously, making a devastating impact on the regularization training relying on steady prediction. For such a challenge, we consider places that always form stable predictions under various disorders are often highly deterministic. Therefore, we propose introducing multi-level disturbances, including noise, iterative, and multi-scale disturbances. We simultaneously present in-depth supervision training on the multi-layer decoder to form stable and consistent multi-scale intermediate predictions, which significantly enrich the number of samples gathered in the following integration process. Then, Monte Carlo sampling is applied to integrate multi-level prediction results to assemble a more robust uncertainty mask map, which gradually clarifies the inter-class feature representation of actual lesions and artifacts. The extended experiment shows that we have successfully and effectively improved the performance of the caries segmentation task by using such uncertain lesion sample features. The dataset is now public, and the code is available at https://github.com/Zzz512/MLUA.

1. Introduction

Caries is the most common oral disease. The World Health Organization has regarded caries, tumors, and cardiovascular diseases as the three crucial diseases for the prevention and treatment of human beings. According to the report of the fourth Chinese oral health epidemiological survey [1–3], there is an extreme gap between caries prevalence rates and treatment rates. The clinical diagnosis of caries mainly relies on panoramic radiographs [4], and the correct interpretation of the panoramic oral image largely depends on the experience of clinical dentists, which heavily increases the uncertainty of caries diagnosis, resulting in relatively high missed diagnosis and misdiagnosis.

Although panoramic X-ray images can provide a comprehensive oral condition and prompt most of the information about dental lesions, there are still the following problems: Firstly, because of X-ray imaging characteristics, panoramic images are always with a lot of noise coming from patients' head swinging, quality of equipment and the operational experience of medical staff, bringing a large number of artifacts in panoramic images and great disturbance to recognition based on low-density shadows. Secondly, different from the general medical segmentation task, the average area of caries lesions only accounts for 1.5‰ of the panoramic

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image, and the shallow caries are even less than 0.5‰. Small target perception is a challenging problem of the modern neural network [5–8]. For images of low concentration with strong noisy background, it is difficult to infer a foreground because the background intensity is much larger than the signal [9]. Finally, caries is a developing disease appearing from the enamel and gradually invading the dentin and pulp cavity, causing it frequently performs a gradual boundary.

There are so many challenges that we need adequate clinical data for research. However, there are still no public panoramic X-ray images for dental caries segmentation. To fill this empty, we introduced a whole novel dental caries dataset called DC1000 (1000 panoramic images of dental caries) for the lesion segmentation task, which consists of over 7500 caries lesions from three different degrees of decay (shallow, middle, and deep caries). The dataset is collected from the Department of Stomatology, Zhejiang Provincial People's Hospital, and assigned to five experienced dentists' joint participation in labeling. As described in the critical problems above, the quality of the images is always influenced by the background noises and disturbances caused by experimental environment and material properties [10], and there are too many interferences make it difficult to draw an exact annotation, so we employ multi-fold cross-validation to selectively screen outlier labels and adjust the blurry areas boundary [11]. Previous experiments show that this operation can improve the performance of the entire dataset and accelerate the convergence of various models without making significant changes to the annotation. We iterate through the above procedures to gradually fine-tune our annotation.

Unfortunately, there are always some controversial areas with relative proportions that cannot be distinguished as lesions or artifacts. All of them show a low-density shadow along with quite different exposure degrees, sizes, and shapes. This means we can neither annotate them nor accurately segment such lesions from panoramic images. To make full use of such kinds of data, we treat them especially as unlabeled data and then introduce a semi-supervised learning framework (SSL). SSL method is mainly used to solve the problem of the insufficient dataset with two mainstream frameworks, including pseudo label supervision [12–14] and consistent regularization [15–18], make use of unlabeled data features and optimize the decision boundary of low-density space. The latter focuses on how to assign a robust uncertainty mask highly sensitive to the semi-supervised training process [19,20], so we associate this specialty with the ambiguous diagnosis of uncertain regions. Given the above challenges, we applied multivariate normal noise, iterative discrepancy, and multi-level predictions for artifact interference, multi-scale, and various morphology issues. The different predictions generated by each particularly designed pathway are treated equally and then integrated into Monte Carlo sampling to determine the deterministic region jointly. We believe those areas still considered as lesions under multiple disturbances are definite lesions to identify those cases that cannot be identified.

Overall, our contributions are summarized as follows:

(1) We propose a whole novel dental caries dataset for the dental caries segmentation task, called DC1000, which consists of 1000 panoramic images with over 7500 caries lesions of 3 classes. To the best of our knowledge, our dataset is the first dental panoramic image dataset and the only public dataset for dental panoramic caries segmentation.
(2) We propose a novel model, Multi-Level Uncertainty Aware (MLUA), that innovatively introduces the uncertainty aware module in the semi-supervised framework to help to differentiate lesions that are difficult to be annotated. Meanwhile, the generation of the uncertain masks, along with the multi-level feature map, makes full use of numerous intermediate predictions and dramatically enhances the scale and morphological consistency representation of the uncertain region with only a little computational cost.
(3) Compared with classic segmentation methods and semi-supervised methods, MLUA makes the best of the characteristics in panoramic images and shows superior performance in several evaluation metrics.

2. Related Works

In this section, we introduce the general studies on computer-aided diagnosis methods for dental caries and semi-supervised segmentation methods in the medical field.

2.1. Computer-Aided Diagnosis for Dental Caries Detection

In diagnosing dental caries, various medical images, including oral panoramic X-ray images, periapical radiography, and bitewing radiography, are usually used by clinical doctors to detect caries. Meanwhile, caries detection tasks consist of caries classification, location detection, and caries segmentation in computer-aided diagnosis.

In the early years, several detection methods are based on traditional image processing or machine learning. In 2017, Singh et al. [21] used a data processing method based on Radon Transformation and Discrete Cosine Transformation for panoramic images to capture low-frequency details. Naam et al. [22] clarified the edges of the dental objects in panoramic images by multiple morphological gradients. This process made the characteristics of proximal caries can be identified precisely. Without extracting high-level semantic information, traditional computer vision processing-based methods obtain a poor generalization performance and thus cannot be competent for complex medical image situations. With the maturity of deep learning since 2016, the method based on the neural network has been on the rise. Ainas et al. [23] utilized pre-trained CNN architecture VGG-16 to classify three major dental diseases, dental caries, periapical infection and periodontitis included. Based on periapical radiography, Lee et al. [24] used a pre-trained CNN model GoogleNet Inception V3 to detect caries at both premolar and molar, showing that CNN algorithms are the most effective methods for detecting caries. Vinayahalingam et al. [25] did a caries classification study only in mandibular and maxillary third molars on panoramic images. Four hundred cropped images from the panoramic images were used to train CNN architecture MobileNet V2 and obtained a high accuracy and sensitivity score. Haihua Zhu et al. [26] proposed a new model called CariesNet to delineate various caries degrees from oral panoramic images. CariesNet, as a U-shape network with the partial encoder module and the full-scale axial attention module, extracts attention features from both the channel domain and the spatial domain helped to segment the caries lesion region more accurately. We summarise the three methods mentioned above as the simple application of deep learning in stomatology. There is no specific design for the natural oral environment, so the application is minimal.

Nowadays, some methods also combine traditional image analysis and deep learning models. Bui et al. [27] combined the features extracted from a 2D tooth image using the deep pre-trained model and the geometric features using mathematical formulas into the new fusion feature, then used a support vector machine as the classifier to classify whether it was caries. Due to the lack of medical image data, there was also a few semi-supervised ways to improve the effect of detection and segmentation. Tuan's group [28] proposed a new caries segmentation framework named Dental Diagnosis System (DDS) based on semi-supervised fuzzy clustering. Besides, the classification task was implemented with a new graphic-based clustering technique. In addition, to improve the detection or segmentation effect, recent research tried to use extra auxiliary tasks with tooth segmentation or tooth layers segmentation. Haghanifar et al. [29] utilized the genetic algorithm and the capsule network as the classifier to isolate teeth in panoramic X-ray images to avoid the influence of several missing teeth. The above method adopts superior algorithms to improve the detection of lesions. Still, their research is based on the assumption of complete perception, ignoring the most important and widespread problem of artifact interference. The detection effect of uncertain lesions needs to be improved.

2.2. Semi-supervised segmentation algorithm in medical domain

Deep learning has achieved great success in image segmentation tasks with increased annotation data. However, it is usually expensive to acquire medical annotation images because generating accurate labels requires expertise and a high cost of worforce. Therefore, the semi-supervised method arises at a historic moment. We can mainly divide current semi-supervised medical image segmentation into three branches [30], including strategy learning with pseudo labels, unsupervised regularization, and knowledge priors prepared. In order to use unlabeled data, the pseudo-label learning method uses the initialization model to generate preliminary pseudo-labels first. It then combines them with the labeled data to update the segmentation model, yielding more accurate pseudo labels. Afterward, the method iterates this step repeatedly to fine-tune a better model. However, the iterative process usually takes plenty of time. Fan et al. [31] adopted a random sampling strategy for enlarging the training dataset with unlabeled COVID-19 images. Specifically, they only used a fixed number of random samples for each training iteration to generate labels. This time-consuming method takes at least a day and a half to finish the whole generation.

Different from the iterative generation of pseudo labels, some studies have been committed to generating supervisory loss based on unsupervised regularization, including consistency learning, co-training, and entropy minimization. Consistent learning is the most widely used among various strategies, enhancing the invariance of input image prediction under different disturbances. Xinkai et al. [32] devised a cross-level contrastive learning method. They proposed a novel consistency constraint that compared the whole image to the prediction of patches to help unlabeled data to enhance the representation ability for local features on two medical segmentation datasets for colon polyps and skin lesions, respectively. This is an all-new consistency alignment method. However, the training is constrained by slicing and overall prediction, which becomes unstable in the environment with a large scale span, so it performs poorly in our work. In addition, some studies used the method of generating pixel-wise uncertain masks to improve semi-supervised disease segmentation diagnosis performance. Lequan et al. [33] designed the uncertainty aware mean teacher model, where the student model can gradually use the Exponential Moving Average (EMA) scheme to learn from the teacher model by both the segmentation loss on the labeled data and the consistency loss against the targets. Specifically, the teacher model adopted Monte Carlo sampling and multiple times Dropout to estimate the uncertainty of each target prediction. The uncertain mask can guide the student model to filter out unreliable noisy predictions and preserve the reliable ones to obtain meaningful targets. This method is highly dependent on the quality of the uncertainty matrix. To get a high-quality uncertainty description, it should improve the sampling efficiency so that the training process can only be carried out in fairly pretty batches. Judge et al. [34] thought current pixel-wise uncertainty estimates were merely based on probabilistic explanation but ignored the knowledge priors. So they proposed a novel contrastive method, encoding the distribution of effective segmentation and then learning the latent space to compare target predictions with thousands of latent vectors to generate an anatomically consistent uncertain map. However, in the case of extremely limited data, the introduction of contrastive learning is hardly effective under the supervision of labeled samples. Therefore, there is only a little work on our panoramic images, which are incredibly prone to artifacts.

Different from general transfer learning [35], the SSL framework no longer requires the teacher network to be a well-trained network to guide the student network. According to earlier work [36] by Nvidia, in the framework of temporary ensembling, two identical networks trained on exactly the same data predict the identical input with quite different disturbances. They expect the two networks should produce as consistent output as possible. However, due to the randomness of the training process, the net will converge in various directions and with high computational costs. Then Tarvainen et al. [37]'s mean teacher thought that it is possible to use the previous experience of one model to learn another and then guide the former, to make its prediction more robust. Here the former is considered a student network, while the latter is a teacher network. Eventually, the teacher network becomes an integrated model, playing the role of AdaBoost or dropout.

3. Methodology

In this section, we introduce our proposed method in detail. Fig. 1 shows the overall architecture of the MULA network. It adopts a twin network structure, including teacher and student networks. First, for the student network, we extract the feature map of each decoder layer as a candidate map for the subsequent uncertain region's decision. Then, to guarantee that the extracted decoder features can provide a confidence map as consistent as possible, we use multiple segmentation heads to train their respective predictions adaptively. For the teacher network, we update the parameters with an EMA to maintain the continuity of its convergence. After passing through the teacher network, the panoramic images mixed with Gaussian noise get direct and multi-layer predictions. All of them are integrated by the Monte Carlo sampling module to generate an uncertainty mask map.

3.1. Problem definition

Given a Dataset $\mathbf{D}$, we divide it into labeled set $\mathbf{D}_l$ and unlabeled set $\mathbf{D}_u$ including M and N samples respectively,i.e. $\mathbf{D} = \{\mathbf{D}_l, \mathbf{D}_u\}$. Then we obtain labeled pairs $\mathbf{D}_l = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{M}$ and unlabeled ones $\mathbf{D}_u = \{\mathbf{x}_i\}_{i=M+1}^{M+N}$. We unify the image sample's dimensions to $C \times H \times W$. We use $\hat{\mathbf{y}}_\phi$ to represent model's direct prediction and $\mathbf{Y} = \{\hat{\mathbf{y}}^l\}_{l=1}^{L}$to represent model's multi-level auxiliary prediction where L stands for extracted number of feature layers. See Table 1 for more details.

3.2. Network Structure

In our MLUA model, we adopt Residual Network (ResNet) as our encoder, and the overall U-shape network structure has been applied as the backbone of the segmentation model. Similar to most semi-supervised models based on the uncertainty aware method [38–41], we employ a twin network architecture including two branches, namely the teacher and the student branch, where the parameters of both share a completely identical structure but with different weights. For student networks, we provide labeled

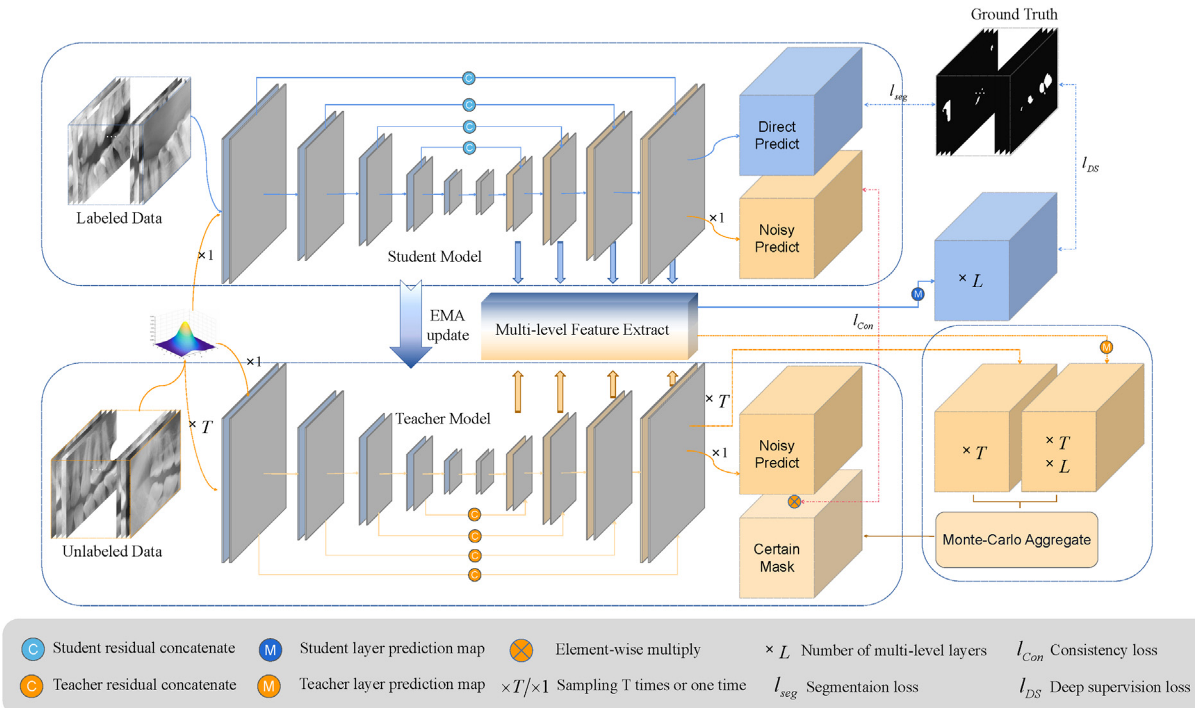

Fig. 1. Workflow of the proposed MLUA model for dental caries segmentation. Two twin networks (teacher and student) are used as the backbone of our task. Firstly, the student network is applied to generate direct predictions with annotated samples. In addition, their multi-level decoder features are also extracted for deep supervision. Secondly, the unlabeled samples with various Gaussian disturbances are transferred into two networks to obtain different noisy predictions. At the same time, the multi-level feature extractor will also produce multiple intermediate forecasts. After that, Monte Carlo is adopted to integrate all sampling predictions to generate a certain mask, according to which consistency constraints are applied between noisy predictions of two networks.

data and then take cross entropy loss and dice loss for training. The definition of a multi-task loss function is as follows:

$$\ell_{seg} = \frac{1}{M}\sum_{i=1}^{M}\left(\ell_{bce}(f(x_i), y_i) + \ell_{dce}(f(x_i), y_i)\right) \tag{1}$$

Where ℓ_{seg}, ℓ_{bce}, and ℓ_{dce} stand for segmentation, binary cross-entropy, and dice loss, respectively. After that, to enable the two networks to provide a model output that is generally consistent but slightly different, we block the backpropagation of the teacher network gradient and updated its parameter weights using EMA as follows:

$$\theta_g = \alpha\theta_g + (1-\alpha)\theta_f \tag{2}$$

therefore, the teacher network parameters update direction is entirely towards the student network but does not become the same. According to temporary ensembling [36], the teacher network integrates student history parameters and becomes the historical version of the student network. In this case, the teacher will produce slightly different predictions but similar to the student network, and the different predictions generated by the identical input under various iterations of the teacher network are defined as iterative disturbances.

Some previous work [42–46] noted the important role of multi-level networks in the scale diversity of various visual tasks, from

Table 1
Main notations and the definitions.

Notation	Definition
D, D_l, D_u	The set of whole samples, labeled samples and unlabeled samples
ϵ, E, γ	Current epoch, max epoch, hyperparameter for adjusting threshold
$f(\cdot), g(\cdot)$	Student and teacher network which takes the input and makes segmentation prediction
M, N, L	The total number of labeled and unlabeled samples, number of decoder layer or upsampling times
M_{decl}, M'_{decl}	Feature map of the l-th decoder and its corresponding feature map with the unified scale after projection
$m_{uncertain}, m_{certain}$	The uncertain mask used to bootstrap the consistency loss, and the certain mask generated by the uncertain mask and a given threshold
$p_\phi(\eta\|x), p(y\|\eta)$	The generated probability of normal distribution given x in the current model ϕ and probability of prediction based on it
$p(y\|\eta^{(t)}), p(y\|\eta^{(t,l)})$	The prediction probability obtained by the t-th sampling and by the l-th layer
R_i, k_i, s_j	Receptive field and convolution kernel size in i-th layer, stride step length of the j-th convolution layer
T	The sampling times by Monte-Carlo aggregate
W_l	The convolution parameters of l-th map layer for segmentation head
x_i, y_i	Input image and its corresponding ground truth of i-th dental caries slice.
$\hat{y}_f, \hat{y}'_f, \hat{y}'_g$	The clean prediction of a student, the noisy prediction of student and teacher
$\hat{y}^l_f, \hat{y}^{(t,l)}_g$	Predicted result of the student l-th decoder layer, and predicted result of the teacher l-th decoder layer at the t-th sampling
$\hat{y}^{mean}_g$	Confidence map integrated by Monte Carlo method with the average of various prediction results
θ_f, θ_g	The parameters of student and teacher network

image pyramids to feature pyramids, selective search to Region Proposal Networks (RPNs), and various variants derived from them. However, they mainly considered how to use multi-level networks to adapt different objects at different scales, ignoring the role of different levels in the detection of identical objects. To successfully guide the uncertainty of feature level, we need to build a multi-level feature extractor, which extracts the features transmitted from each layer decoder to the next, denoted as $\{\mathbf{M}_{dec1}, \mathbf{M}_{dec2}, \ldots, \mathbf{M}_{decl}\}$, where $\mathbf{M}_{decl} \in R^{C_l \times H_l \times W_l}$.In order to unify different level feature maps, we apply a simple 3×3 convolution followed by a Relu operation and then upsample them to the feature map of the unified shape $C_L \times H_L \times W_L$ according to the final decoder layer. Now we have $\{\mathbf{M}'_{dec1}, \mathbf{M}'_{dec2}, \ldots, \mathbf{M}'_{decl}\}$ where $\mathbf{M}'_{decl} \in R^{C_L \times H_L \times W_L}$.

3.3. Multi-level Monte-Carlo Aggregate

Deep supervision is a simple but effective constraint method, which forces each decoder layer to fit the ground truth, and then the model could generate prediction results with a higher degree of confidence [47]. According to the role of receptive field in neural network architecture based on CNN [48], now we consider that these prediction results will fairly participate in the following description of the uncertain region, it needs to rigorously avoid the over shrink of small targets in the deep encoder or decoder stage. We have the receptive field of each step according to the following formula:

$$R_i = R_{i-1} + (k_i - 1) \cdot \prod_{j=1}^{i} s_j \tag{3}$$

where R_i and k_i represent the size of the receptive field and kernel size at the layer of i respectively, and s_j means the convolution stride length accumulated to the j step. We carefully adjust the sampling times to ensure that most of the small caries lesions do not drop in the situation of being off the track. Otherwise, they would become uncertain regions in perpetuity and never contribute to semi-supervised consistency training, which is disastrous for improving the detection of small caries. We use L segmentation heads to train our multi-level extractor with the loss function of deep supervision as follows:

$$\begin{aligned} \ell_{DS} &= \tfrac{1}{M}\tfrac{1}{L}\ell_{bce}\left(\sigma\left(\mathbf{W}_l * \mathbf{M}'_{(decl,i)}\right), \mathbf{y}_i\right) \\ &+ \ell_{dce}\left(\sigma\left(\mathbf{W}_l * \mathbf{M}'_{(decl,i)}\right), \mathbf{y}_i\right) \end{aligned} \tag{4}$$

where M and L stand for labeled samples extracted layers number.$\mathbf{W}_l$ and δ denote weight matrix of l layer and sigmoid function respectively. We conduct deep supervision training on the student network through labeled data and then deliver it to the teacher through EMA to be capable of making paralleled predictions.

In the semi-supervised framework, the Monte Carlo method is generally used to simulate and sample to estimate the distribution of target characteristics [49]. In order to obtain robust sampling results, some works [50,39] add gaussian noise to medical images to generate differential predictions and then collect the results of the input image on a specific model. As described in the paper, given the input $\mathbf{x}$, we let η be a certain multi-dimensional normal distribution in the feature space generated by $\mathbf{x}$. We use the following Monte Carlo approximation to get this distribution:

$$p(\mathbf{y}|\mathbf{x}) = \int p(\mathbf{y}|\eta)p_\phi(\eta|\mathbf{x})d\eta \approx \frac{1}{T}\sum_{t=1}^{T} p(\mathbf{y}|\eta^{\mathbf{t}}) \tag{5}$$

where $\eta|\mathbf{x} \sim N(\mu(\mathbf{x}), \delta(\mathbf{x})), \mu(\mathbf{x}) \in R^{C \times H \times W}$ and $\delta(\mathbf{x}) \in R^{(C \times H \times W)^2}$. Considering that the direct prediction $\hat{\mathbf{y}}_f$ of the model and the multi-level prediction $\hat{\mathbf{y}}_f^l$ adopted in this paper trained with exactly the same weights, we regard them as independent prediction units. Thus we can expand the simulation process generated by the above formula:

$$p(\mathbf{y}|\mathbf{x}) \approx \frac{1}{T}\frac{1}{L}\sum_{t=1}^{T}\sum_{l=1}^{L} p\left(\mathbf{y}|\eta^{(t,l)}\right) \tag{6}$$

Suppose we have collected all direct and multi-level predictions denoted as $\hat{\mathbf{y}}_g^{(t,l)}$ where $l \in \{0, 1, \ldots, L\}$. We treat $l = 0$ as the direct segmentation prediction of the model and others corresponding to multi-level outputs, respectively.

Now we will describe the implementation process of Monte Carlo sampling and integration. It should be noted that only unlabeled data will be put into the operation. For a dental panoramic image slice under test, we need to apply $2 + T$ times of independent gaussian noise for it to generate noisy samples with a slight difference. Two are input into the student and teacher networks, respectively, as a reference prediction for consistency loss training, and the remaining T are all transferred to the teacher networks for sampling. Then we have collected $T \times (L + 1)$ various outputs. Compared with the models of other consistent regularization methods [39], we obtain extra multiple prediction results with only a small amount of computation (additional segmentation heads). Under the description of the above formula, these supplementary prediction maps have equal contributions with direct prediction referred to "(6)".

Now we will integrate all predictions obtained above in the way shown in Fig. 2. First, we will collect all the teachers' outputs, totaling $T \times (L + 1)$, which are sequentially spliced into a tensor of $T \times (L + 1) \times C \times H \times W$. Then it is pooled into a $C \times H \times W$ matrix by the mean pooling, denoted as $\hat{\mathbf{y}}_g^{mean}$, and the uncertain mask map is obtained by the following formula:

$$\mathbf{m}_{uncertain} = -2\hat{\mathbf{y}}_g^{mean} log(\hat{\mathbf{y}}_g^{mean}) \tag{7}$$

where we multiply by two because, unlike the work in [39], our prediction logits are in binary sigmoid form, without which the following threshold will be permanently invalid due to the maximum of $\mathbf{m}_{uncertain}$ always being smaller than the minimum of the threshold by "(8)". Now we assign to pick out a certain field to make consistent regularization more stable. Considering that the prediction of the model along with the training becomes gradually reliable, the dynamic threshold is generally used to adaptively increase the range of the deterministic region, which is determined as follows:

$$threshold = \gamma \cdot (\beta + (1 - \beta) \cdot e^{-5 \cdot (1 - \frac{\epsilon}{E})^2}) \tag{8}$$

where γ and β is two hyperparameters for adjusting threshold, ϵ and E refer to current and max training epoch respectively. The part of the uncertainty measurement map smaller than the current training threshold is selected as the certain mask $m_{certain}$.

3.4. Uncertainty Aware Training

Now consistent regularization loss training is used to establish the link of unsupervised samples in our twin network. We in conformity with the deterministic mask to select certain regions from student and teacher network's direct prediction $\hat{y}'_f$ and $\hat{y}'_g$. Then we apply L2 mean square error (MSE) loss for unsupervised training, the loss function is designed as follows:

$$\ell_{con} = \frac{1}{N}\sum_{M+1}^{M+N} M_{certain} \cdot (\hat{y}'_g - \hat{y}'_f)^2 \tag{9}$$

finally, we define the loss function of the whole training process as:

$$\ell = \ell_{seg} + \ell_{DS} + \lambda\ell_{con} \tag{10}$$

where λ is a balanced factor for tradeoff loss function between labeled and unlabeled training process. Here, the value of λ follows the sigmoid ramp-up policy [36], which is defined as follows:

$$\lambda = w \cdot e^{-5\cdot(1-\frac{\epsilon}{E})^2} \tag{11}$$

similarly, ϵ and E represent the current and max epoch, respectively. At the same time, w takes a relatively small fixed value to ensure that the loss will not yield trouble to the student network when the early training is unstable. Eventually, we put all the data flow in pseudocode 1.

Algorithm 1: Pseudo Code for proposed MLUA

Input: labeled data $\mathbf{D}_l$, unlabeled data $\mathbf{D}_u$
Output: student model $\mathbf{f}(\cdot)$
initialize student and teacher parameters $\mathbf{f}$, $\mathbf{g}$;
sampling $(\mathbf{x}_i, \mathbf{y}_i)$ from $\mathbf{D}_l$, $\mathbf{x}_j$ from $\mathbf{D}_u$;
while *no convergence* **do**
    $t \leftarrow 0$;
    $\hat{\mathbf{y}}_f, \hat{\mathbf{y}}_f^l \leftarrow \mathbf{f}(\mathbf{x}_i)$;
    obtain segmentation loss ℓ_{seg} by $\hat{\mathbf{y}}_f$ and $\mathbf{y}_i$;
    obtain deep supervised loss ℓ_{DS} by $\hat{\mathbf{y}}_f^l$;
    $\hat{\mathbf{y}}_f', \_ \leftarrow \mathbf{f}(\mathbf{x}_j)$;
    $\hat{\mathbf{y}}_g', \_ \leftarrow \mathbf{g}(\mathbf{x}_j + noise)$;
    $volume \leftarrow zeros\_like(T \cdot L, C, H, W)$;
    for $t < T$ **do**
        $\hat{\mathbf{y}}_g^t, \hat{\mathbf{y}}_g^{(t,l)} \leftarrow \mathbf{g}(\mathbf{x}_j + noise_t)$;
        $volume$ concatenate $\hat{\mathbf{y}}_g^t, \hat{\mathbf{y}}_g^{(t,l)}$;
        $t \leftarrow t + 1$;
    end
    $\hat{\mathbf{y}}_g^{mean} \leftarrow meanpooling(volume)$;
    obtain $\mathbf{m}_{uncertain}$ through eq(7);
    $\mathbf{m}_{certain} \leftarrow \mathbf{m}_{uncertain} < thresh$;
    obtain consistency dist by $\hat{\mathbf{y}}_f'$ and $\hat{\mathbf{y}}_g'$;
    $\ell_{Con} \leftarrow mean(\mathbf{m}_{uncertain} \cdot dist)$;
    update student $\mathbf{f}$ by ℓ_{seg}, ℓ_{DS}, and ℓ_{Con};
    update teacher $\mathbf{g}$ through eq(2);
end

4. Proposed Dataset

Due to the lacking of the Carious lesion Segmentation dataset based on Oral Panoramic images mentioned in the first Chapter, we collect and annotate a new middle-scale carious lesion dataset named DC1000, which contains 1000 Panoramic X-ray images divided into 593 detailed annotation images and 407 rough annotation images. In this section, we present the process of panoramic image acquisition, illustrate the cleaning of annotated data, and the detailed attribution of this proposed dataset. In our final cleaned dataset, the whole oral image is consistent with the clinical circumstance in the hospital, with a width of 2943 pixels and a height of 1435 pixels. The following is an introduction(refer to [51]) to data properties with 593 fully labeled panoramas.

Carious count. After statistics, there are 4520 caries in fully labeled oral panoramic images totally and equally 7.6 caries per panoramic images. In the outside ring of Fig. 3(a), we illustrate the data distribution of dental caries count in all patient oral

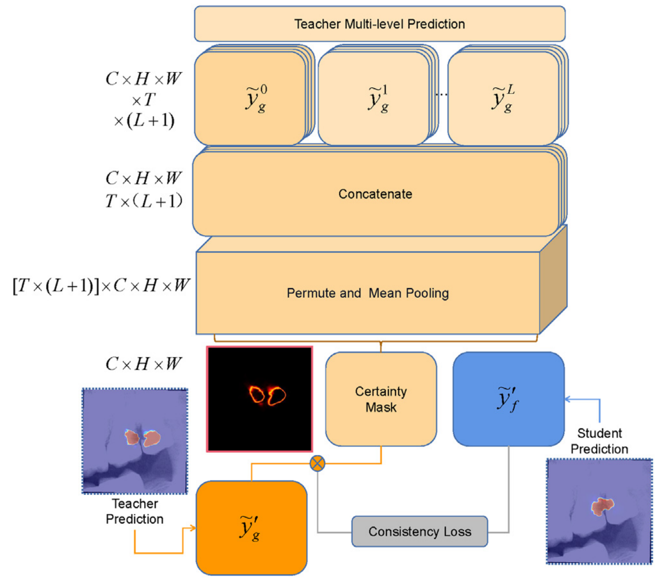

Fig. 2. Details of the Monte Carlo aggregation. Direct and multi-level prediction with diverse noise extracted from the teacher network is concatenated together for makeup $T \times (L+1)$ sampling. After that, each sample mean pooled becomes a certain mask according to a dynamic threshold. The final certainty mask guides the consistency loss of teacher and student predictions. The heatmap with a black background in the middle represents the uncertainty mask, and the higher temperature of the mask refers to higher uncertainty.

panoramic images. We observe that approximately 77% of patients have caries less than 11 in our labeled dataset. Moreover, nearly 6% of patients have more than 15 caries. This dataset is consistent with routine clinical care naturally.

Foreground proportion. Due to the small region of caries, it is also very hard and strenuous for doctors to locate dental caries which introduced in first Chapter. In the upper right corner of Fig. 3(b). shows the foreground proportion in all patients' panoramic images. Approximately 73.5% of images hold the caries region ratio of less than 1.5‰, namely the mean total caries region of all panoramic images is about 6335 pixels which are far from the whole radiograph with the area of 4223205 pixels (2945×1435). However, in most medical image segmentation tasks, such as the Colon Polyp Segmentation task which intestinal polyp's area proportion is about 7 percent on average. In order to increase foreground percentages and enhance the ability to extract features from network models, we crop the entire training dataset into 384 square areas so that our average foreground area can reach 11.79%. Consequently, we obtain 1460 training slices to improve our annotation quality.

Dental caries area distribution. Since the neural network has distinct difficulties in identifying the small and large areas of dental caries, we had better count the distribution of different carious lesion sizes. The carious lesion region is divided into nine parts based on lesion area. The specific division interval results and the corresponding number of caries are shown in the low left corner of Fig. 3(b). We can observe that nearly three-quarters complete caries' area is under 500 pixels, especially the first two intervals. It means that our general task is mainly focused on small and medium size caries.

Category count. According to the property of dental caries, all caries lesions in our dataset are divided into four level classes, including shallow caries, middle caries, and deep caries. In the inside and middle ring of 3(a), we illustrate the data distribution of various types of dental caries count in all patients' oral panoramic X-ray images. The data distribution in our dataset is balanced with the approximate quantity of three kinds of caries. However,

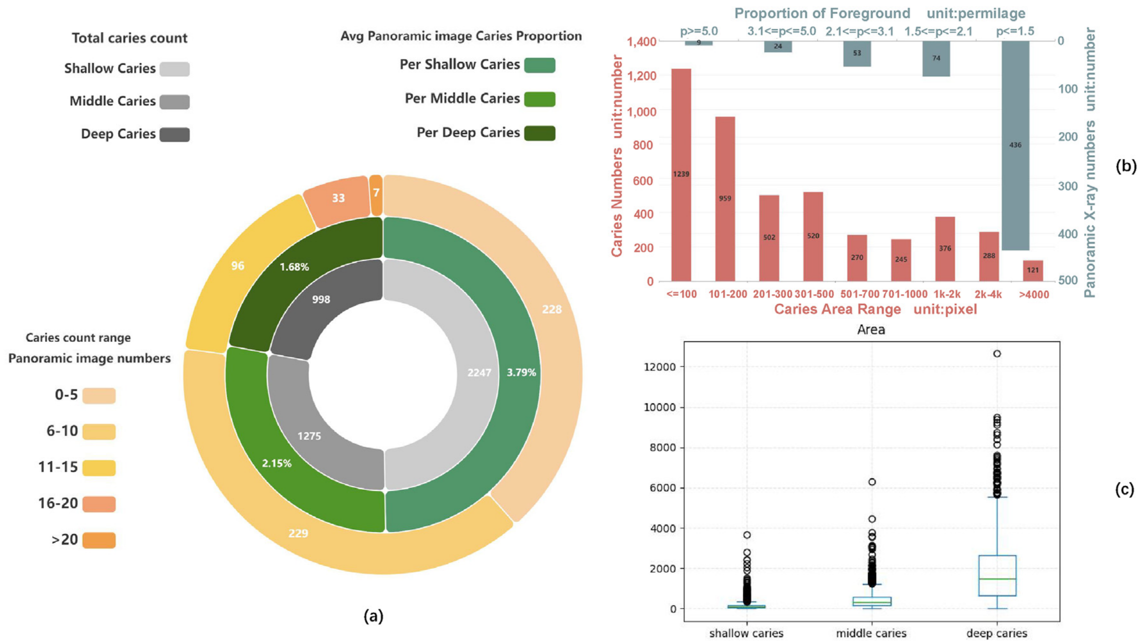

Fig. 3. Here are some of the main characteristics of our collected dataset, including caries count information in panoramic images (a), relevant information of caries regions area (b), and the box plot of different caries regions area (c).

we observe that the area of caries in each category is irregular in Fig. 3(c). The proportion of shallow caries in the total area is quite small, while deep caries accounts for the vast majority. In this picture, the box describes the general distribution of each kind, and the black circle represents outliers (but correct data). The average pixel area in these three types of caries is 160,474,1952, respectively. The size of shallow and medium caries is relatively small, while that of deep caries is rather large, so it will be more challenging to identify shallow or middle caries.

5. Experiments

In this section, we briefly describe the content of our experiment, starting with the evaluation metrics we used and some experimental setting details. The classical segmentation network is then introduced to our dataset as a benchmark model to evaluate our dataset. For fairness, we introduced some semi-supervised models based on uncertainty aware consistency regularization for performance and efficiency comparison. We added some ablation experiments to verify the effectiveness of our proposed model in the complex segmentation task of panoramic caries. Finally, we demonstrate some visual analysis to examine the effect of caries detection.

5.1. Evaluation Metrics

We selected the most commonly used segmentation evaluation metric including accuracy, dice score, precision, sensitivity and specificity. Because of the particularity of the task, the background accounts for too much in the panoramic images and the model can often correctly predict true negatives, which makes the accuracy and specificity of our dataset higher than 99%. Thus we remove these two meaningless evaluation indicators. We conducted fully supervised and semi-supervised performance evaluation under 265 and 530 slices in two groups of experiments, respectively, denoted as $DICE_{265}, SEN_{265}, PRE_{265}$ and $DICE_{530}, SEN_{530}, PRE_{530}$. Finally 100 panoramic X-ray images in the data setwere selected as evaluation data.

5.2. Implementation Details

We train our model on PyTorch Lightning with a NVIDIA GeForce RTX 3090 GPU. All segmentation models adopt ResNet34 as encoder including twin networks in semi-supervised and single network in fully supervised. In our MLUA model, we use the AdamW optimizer with learning rate of 0.001, the momentum and wight decay coefficients is set to 0.9 and 0.001, respectively. And the batchsizes of full supervision and semi-supervision are set to 4 and 8, respectively. Among the 8 batchsizes of semi-supervision, labeled and unlabeled samples account for half of each other. In addition, we take the maximum number of epochs as 200, and set the number of layers of multi-level decoder as 4, that is $E = 200$ and $L = 4$. Besides, $\alpha = 0.99, \beta = 0.75$, and $w = 0.1$ are set for the remaining three hyperparameters mentioned in the article.

5.3. Baseline and Comparison

In this section, we first introduce the baseline built on our DC1000 dataset using some classic image segmentation networks. The implementation models include MANet, FPN, DeepLabV3+, Linknet, Unet, Unet++, PAN, and PSPnet. In order to make a fair comparison, we also introduce some commonly used semi-supervised learning frameworks to test our targeted solutions' effectiveness for identifying controversial areas in the caries segmentation task. The results are shown in Table 2. The overall DICE performance of our dataset in all methods besides ours is below 70%. And the sensitivity of each model is less than precision to varying degrees, which is caused by the small proportion of caries area in the panoramic image. Under fully supervised training with 265 slices, the networks have significant performance differences. The worst performance of MAnet is 40.37%, and the best is that of PSPnet, 54.43%. When we double the number of training slices to 530 pieces, their performance is stable at 65% DICE, which shows the stability of the proposed dataset. Here we focus on the classical segmentation network Unet and the backbone model FPN, their segmentation accuracy rates are 64.99% and 64.66%, respectively.

Table 2
Performance comparison of classic segmentation network and uncertainty aware based semi-supervised model.

Method	$DICE_{265}$	SEN_{265}	PRE_{265}	$DICE_{530}$	SEN_{530}	PRE_{530}
MAnet	40.37	39.37	54.81	65.03	61.06	75.34
FPN	43.43	41.90	55.62	64.66	59.30	76.30
DeepLabV3+	46.94	40.75	68.07	61.23	52.72	**82.01**
Linknet	45.10	47.11	53.50	66.08	61.47	77.07
Unet	42.51	43.68	53.89	64.99	63.07	72.74
Unet++	45.63	41.73	60.84	64.09	60.44	74.10
PAN	46.46	43.54	60.63	65.60	60.46	76.72
PSPnet	54.43	46.73	**76.36**	65.11	59.01	78.64
URPC	49.44	43.74	71.78	59.34	51.77	81.17
UAMT	52.31	52.90	59.14	67.79	64.60	76.02
CLCC	56.23	52.07	67.76	65.55	58.48	79.46
Ours	**61.40**	**58.77**	70.04	**71.12**	**68.44**	76.94

Next, we selected some semi-supervised models that are partially similar to our methods, including URPC, UAMT and CLCC. Their main characteristics and the difference between us are described as follows:

1) **Uncertainty Rectified Pyramid Consistency(URPC)**: A backbone is extended to produce pyramid predictions in different scales. The pyramid predictions is supervised by the ground truth of labeled images, and the loss of multi-scale consistency is guided by unlabeled images to ensure the consistency of the identical input at different scales. We are the same in implementing the multi-scale feature predictions but distinguished in that we take the twin network for consistent regularization instead of doing this between different scales.
2) **Uncertainty Aware Mean Teacher(UAMT)**: The most commonly used twin network for the semi-supervision framework consists of a student model and a teacher model. The student learns from the teacher by minimizing the segmentation loss and the loss of consistency about the target area. We expand the main components of the framework to form an uncertain joint map from EMA and noise sampling to multi-level predictions and modify the established relationship between the certain mask and the dynamic threshold.
3) **Cross-level Contrastive Learning and Consistency Constraint(CLCC)**: Because the information contained in the complete medical image and the image slice is identical, the results of the features pieced back after cutting should be consistent with the reasoning results of the original image features. After slice transformation, the original image and the slice should reveal the same representation as much as possible within both the feature level and image level so that the representation of the unlabeled image can be learned without additional annotation. This is another attempt at consistency constraint. The main difference between our work and CLCC is that we take the feature representation to build a deterministic mask and then only use image prediction to generate consistency loss.

Among the three models mentioned above, the performance of URPC is the worst, 49.44% and 59.34% are compared with our 61.40% and 71.12%. In the remaining two semi-supervised networks, UMAT and CLCC, we can see that they have better performance when the labeling ratio is small (265 slices). However, under the semi-supervised training condition of 530 slices, their performance is limited compared with that of the fully supervised network. The above results mean that introducing SSL can effectively improve the problem of a large number of uncertain regions in our tasks. However, the improvement of these methods is limited. In contrast, we introduce iterative, noisy, and multi-scale disturbances to jointly participate in the decision-making in the uncertain region and successfully improve the best performance to 61.40% and 71.12%(5.17% from 56.23% and 5.04% from 66.08%), respectively. Still, for such a diagnostic environment, higher sensitivity is more beneficial to early diagnosis. We enumerated the effects of partially fully supervised and semi-supervised models as shown in Fig. 4.

5.4. Ablation Study

In order to verify the effectiveness of the proposed method, we set up eight ablation experiments to test the three various disturbances. We define respectively the historical difference generated by EMA update as iterative disturbance p_{iter}, the input difference generated by multivariate normal distribution as noisy disturbance p_{nois}, and the prediction difference generated by multi-level as multi-scale disturbance p_{ms}. The control conditions of various disturbances are as follows:

(1) When p_{iter} is introduced, the teacher network enables the EMA update mode, otherwise it directly inherits all the parameter weights from the student network at the end of each epoch.
(2) When p_{nois} is introduced, we add the random number generated by multivariate gaussian normal distribution to the input, otherwise we do not take any process.
(3) When p_{ms} is introduced, we also enable deep supervision and incorporate multi-layer predictions into Monte Carlo, otherwise only the direct prediction of the last output of the segmentation network is sampled.

The results of the ablation experiment are shown in Table 3, which reveal the effectiveness clearly of introducing various disturbances to improve performance. First of all, when separately introducing three different disturbances the performance can reach 56% from 48% and 67% from 64.66%, respectively. When we introduce iterative disturbance and noisy disturbance simultaneously, it's $DICE_{530}$ increases to 68.02% (3.36% from the baseline). Then, when we combine either of the first two kinds of disturbances with our specially designed multi-scale disturbance combination, we obtain 59.65% (69.00%) and 59.38% (69.52%), improved by 11.80% (4.34%) and 10.81% (4.86%) from the baseline, respectively). At last, when we take all the disturbance information, we achieve 61.40% for $DICE_{265}$ and 71.12% for $DICE_{530}$, which increases the sensitivity from 44.65% (59.30%) to 58.77% (68.44%).

Compared with the change tendency of SEN and PRE in Table 3, with the increase of the SEN, the PRE began to decrease, which means that the missed detection case was improved by our method. Still, the model introduced some wrong detections. As a result, our final PRE_{530} did not perform as well as without p_{ms} but had a better overall performance (dice score). For medical image detection tasks, the consequences of a missed detection always lead to more serious consequences than a false detection, so such results are entirely acceptable.

At the same time, to verify the proposed method's ability to deal with multi-scale challenges, we classify the histogram shown in the low left corner of Fig. 3(b) into one class every three clusters, corresponding to small groups with less than 300 pixels, medium clusters with more than 300 pixels and less than 1000 pixels, and large groups with more than 1000 pixels. Table 4 shows the corresponding results. We have excellent performance on both $DICE_m$ and $DICE_l$ except for $DICE_s$. For the problem that UAMT performs better than ours on small-scale objects segmentation, we consider that because we sampled four times as many times as the UAMT to participate in the generation of the uncertain map

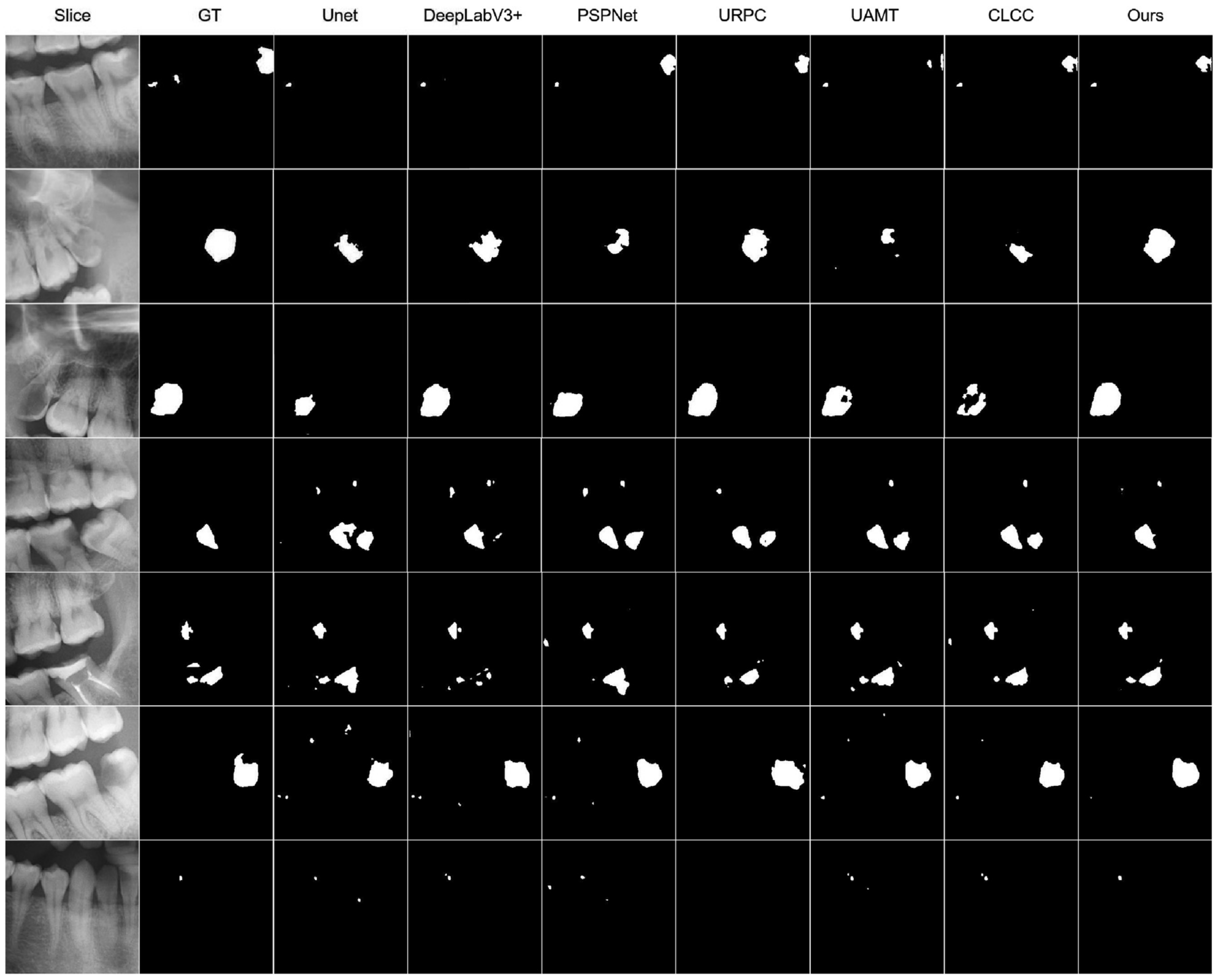

Fig. 4. For quality comparison between different models, we imaged the results in slices for convenience of presentation. Where Unet, DeepLabV3Plus, and PSPNet displayed in columns three to five are fully supervised models, while URPC, UAMT, CLCC, and our method are semi-supervised models shown in the rest of the columns.

Table 3
Ablation experiments under three different perturbation combinations.

p_{iter}	p_{nois}	p_{ms}	$DICE_{265}$	SEN_{265}	PRE_{265}	$DICE_{530}$	SEN_{530}	PRE_{530}
			48.57	44.65	63.56	64.66	59.30	76.30
✓			56.19	53.18	67.54	67.00	64.71	72.97
	✓		51.78	54.33	57.50	67.60	63.83	76.92
		✓	56.95	51.66	71.61	67.30	62.52	77.03
✓	✓		55.61	51.41	69.16	68.02	62.63	79.45
✓		✓	59.65	55.05	71.02	69.00	66.13	75.52
	✓	✓	59.38	56.66	68.48	69.52	67.56	74.74
✓	✓	✓	61.40	58.77	70.04	71.12	68.44	76.94

during the consistency alignment phase. The uncertain map formed by the larger fluctuation of the small-scale target area during multiple predictions becomes more unstable, resulting in $DICE_s$ not as good as that of UAMT.

Table 4
Comparison of multi-scales between different methods.

method	$DICE_s$	$DICE_m$	$DICE_l$
DeepLabV3+	14.47	29.65	63.35
Unet	17.67	35.13	66.85
CLCC	17.80	43.16	66.68
UAMT	24.39	40.94	73.70
Ours	20.16	**47.46**	**75.81**

To further explore the advantages of the proposed method, as shown in Fig. 5, we find that the speed of the proposed model also becomes faster under the same sample conditions. When batchsize is unified to 8, and the number of samples was set to 5, the average iteration number of our model reached 25.97 times per second, which was 1.17 times the speed of UAMT's 22.27. When the sampling number was set to 160, the average iterations per second were 2.37 and 0.99, respectively, meaning the proposed model

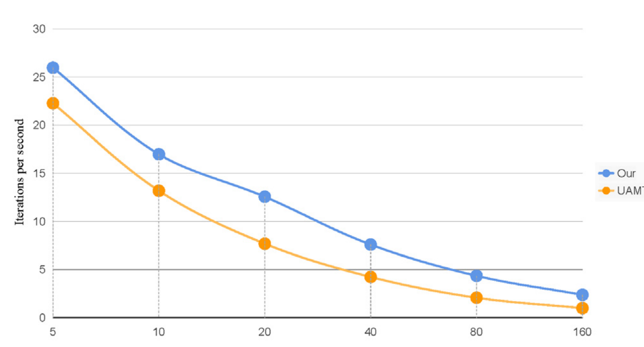

Fig. 5. Here is the model efficiency comparison between UAMT and the proposed method. The proposed method has obvious speed advantages under the condition of the same number of sampling.

Table 5
Ablation experiments under three different perturbation combinations.

T	S_n	$DICE_{530}$	SEN_{530}	PRE_{530}
1	5	69.13	67.74	73.81
2	10	70.06	66.40	76.73
4	20	70.56	62.52	67.24
8	40	71.12	68.44	**76.94**
16	80	69.27	**74.21**	67.83

was 2.39 times faster than that of UAMT. We attribute this to the involvement of multi-level in predicting the uncertainty mask.

To make a profound study for the upper limit of the proposed method, we set up a tentative experiment shown in Table 5 with different sampling conditions. We keep the number of multi-layer feature extraction as ($L = 4$), hence each time we conduct a sampling, we have five predictions, including one direct prediction and four intermediate predictions. Under all different configurations, our method can achieve more than 69% performance. With the increase of sampling numbers, the model's performance could also improve and achieve the best performance when sample number $S_n = 40$.

6. Conclusion

In this paper, we have proposed a novel dataset based on dental panoramic X-ray images for the caries segmentation task. Due to the multi-scale distribution with imbalanced proportions between foreground and background, diversity of shapes, and the existence of artifacts that are even hard for doctors to distinguish, this task is more complicated and challenging than other general medical detection missions. We introduce a semi-supervised method to help gradually identify region areas that are uncertain of segmentation. Moreover, we propose consistent multi-level regularization to enrich the fuzzy field's prediction confidence, improving their recognition performance for various scale caries. The extended experiments show that our method successfully improves the accuracy of the segmentation network, and reaches a new performance among several semi-supervised frameworks. However, the existing experimental data are still far from reaching the standard of human detection accuracy. In our future work, we focus on how to strictly classify the detected caries and expand our dataset capacity (mainly unlabeled data) to improve the overall detection ability to play a clinical auxiliary role for dentists.

CRediT authorship contribution statement

Xianyun Wang: Conceptualization, Methodology, Software. **Sizhe Gao:** Formal analysis, Data curation, Investigation. **Kaisheng Jiang:** Methodology, Software, Writing - original draft. **Huicong Zhang:** Validation, Resources, Investigation. **Linhong Wang:** Software, Visualization. **Feng Chen:** Writing - review & editing. **Jun Yu:** Resources, Supervision, Writing - review & editing. **Fan Yang:** Supervision, Project administration.

Data availability

The authors do not have permission to share data.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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